Zhiwei Li

AI Full Stack Engineer

Guangzhou, China

(+86) 132-6163-0925 | lzwjava@gmail.com | lzwjava.com | lzwjava | lzwjava



Truth

Summary

- Male, born in 1995, Chinese citizen, resides in Guangzhou. Achieved top 300 in Guangdong NOIP 2011 and solved around 1000 algorithm problems online.
- Studied for one year at Beijing Forestry University before dropping out. Pursuing an associate degree through self-taught education, with 13 written exams and 3 practical exams passed. Pursuing a Master of Science in Computer Systems and Engineering (MSCSE) at California Science and Technology University, currently ongoing.
- Currently open for new opportunities. Previous experience includes TEKsystems (outsourced to HSBC Bank), LeanCloud, Beijing Square Root, Beyondsoft (outsourced to DBS Bank), and Farben Information (outsourced to HSBC Bank).
- Accumulated 12 years of professional experience, including 9 years in corporate roles and 3 years as a freelancer. Specializes in 2 years of mobile (Android, iOS) engineering, 8 years of full-stack engineering (7 years backend, 1 year frontend), and 2 years in machine learning/big data.
- Good understanding of HTML, CSS, Browser APIs, JavaScript, Vue.js, React, Angular, and iOS/Android development. Familiar with Java, Spring, Spring Boot, MySQL, Redis, APIs, networking, containers, and distributed systems. Familiar with logging, database, and microservice modules on Alibaba Cloud, AWS, and Azure.
- Native in Chinese, proficient in English (IELTS 6.5). Read over 320 books spanning textbooks, personal development, business, technology and history. Read around 8000 comprehensive, long AI answers in the past year.
- Developed 10 open-source GitHub projects as a main developer, each with over 500 commits available at https://github.com/lzwjava. Detailed software portfolio available at https://lzwjava.com/portfolio-en.
- Generous knowledge share with around 400 original posts on the blog attracting around 75,000 page views a month, and who documents a lot in the corporate environment too.
- Self-taught researcher with an academic-style paper on reversing myopia and natural vision restoration, inspired by the works of Yin Wang and Todd Becker, backed by three years of experimentation.
- AI Enthusiast — Trained 13 small LLMs (nanoGPT, GPT-2 124M/760M/1.5B) on H200, AMD MI300X and RTX 4070 GPUs, built SEC-EDGAR-GPT from scratch on SEC filings, and built personal AI projects using OpenRouter and Claude Code (~5B tokens used over the past year).

Work Experience

TEKsystems *Guangzhou, China*

Full Stack Engineer (Outsourced to HSBC Bank) *Feb 2025 - Jul 2026*

- Full Stack Engineer outsourced to HSBC Bank under TEKsystems, leveraging Copilot and its API to accelerate backend development for a module of the Finance Transformation Platform in HSBC's Enterprise Technology Department. The module covers finance adjustments in the finance ledger platform, managing journal-level controls for adjustments and finance journals.
- Maintained features for financial data processing including import, validation, and export, while enhancing submission and approval workflows for new financial headers. Gained experience in accounting, ledger, and banking systems.
- Participated in the full development lifecycle, from local development to UAT testing and production releases. Helped decommission legacy applications running on WebSphere Application Server. Automated releases using Ansible and Jenkins, assisted in upgrading Angular to a major version, and enjoyed tackling all kinds of tasks—especially resolving bottlenecks for the team.
- Led integration and API testing for the module, using the Copilot API to auto-generate around 70 test cases covering Spring Filters in Java, Python unittest, JSON truncation, prompt engineering, and regional endpoints.
- Made myself an autonomous humanoid AI agent to automate as much as possible—through scripting, logging, documenting, and sharing. By implementing caching and validation of LLM outputs, creating dedicated prompts and contexts for 20 customized agents, and leveraging 400 reusable Copilot-written scripts and 1,100 Copilot-written guides across the tech stack, the engineering experience in a large bank has started to feel good. And I ranked top 6% in Copilot usage internally measured by premium requests metric. Joined HSBC's internal AI community and earned a Contribution Award for the high-profile AIPlayer project.
- Worked with technologies such as Java, Spring, IBM Db2, Maven, Nexus, Angular, Python, HashiCorp Vault, Ansible, along with tools including Windows, Control-M, IBM WebSphere Liberty Profile, and Copilot.
- Worked with a Principal Engineer on a project involving large Excel file handling and a join/union tool to compare gaps and differences across multiple banking systems, using Python, DuckDB, React, and Copilot.
- Continuously explored AI after work by training small models (nanoGPT, GPT-2 760M, SEC-EDGAR-GPT 124M) and fine-tuning models on personal knowledge base text (~5M tokens) across H200, RTX 4070, and AMD MI300X GPUs; running inference with Qwen-27B; building personal projects using OpenRouter and Claude Code (~5B tokens consumed in the past year); and learning the fundamentals of PyTorch, ROCm, FlashAttention, Codex, llama.cpp, Transformers, and reasoning techniques. Joined the NVIDIA Accelerated Computing Developer Tech Exchange and learned techniques from industry experts.

Self-Employed

Guangzhou, China

Independent AI Engineer

Aug. 2023 - Jan. 2025

- Analyzed and reimplemented core components of approximately 30 machine learning projects from PyTorch, TensorFlow tutorials, Coursera courses, and open-source repositories. Earned two Coursera certificates: Machine Learning Specialization and Deep Learning Specialization.
- Prepared for exams to obtain an associate degree, focusing on advanced mathematics, computer organization, and linear algebra. Studied Japanese through immersion in Japanese music and TikTok videos.
- Contributed as a Full Stack Developer to an AI-powered story bot leveraging Claude's API for personalized storytelling. Implemented prompt configuration and a management interface using Python, Flask, React, and Nginx, with deployment on AWS. Employed Prometheus for monitoring, ELK stack for log management, and ChatGPT-4 for coding support.
- Maintained a technical blog with 431 original posts at https://lzwjava.com. Utilized large language models for translation into 9 languages, text-to-speech audio, and support for PDF and EPUB formats. Integrated GitHub workflows and used LaTeX for academic papers and resume creation. Technologies included Python, Jekyll, Deepseek, and Mistral.
- Conducted experiments with llama.cpp, embeddings, rerankers, Retrieval-Augmented Generation, and the MMLU benchmark. Integrated search engine APIs, including Jina AI, Tavily AI Search API, and ElevenLabs API.

Farben Information

Guangzhou, China

Backend Engineer (Outsourced to HSBC Bank)

Nov. 2022 - Jul. 2023

- Farben is a leading software technology service provider in China. HSBC Bank is one of the largest banking and financial services institutions in the world. PayMe is an HSBC mobile payment service catering to Hong Kong residents, serving more than 3 million users.
- Contracting through Farben, assigned to work for HSBC Bank and contributed to the PayMe project. Participated in the backend development for the Auto Top Up feature, which automatically tops up funds from the user's credit or debit card when their balance falls below a certain amount. Monitored post-payment events from Azure EventHub and checked for top-ups whenever a user set up their auto top-up configuration. Used dedicated object-oriented programming to handle cases elegantly and applied aspect-oriented programming to audit logs for changes in the auto top-up table.
- Actively contributed to cloud migration following participation in the company's AWS training. Refactored APIs to utilize request header-based routing, ensured secure access and database configurations, and participated in deploying microservices to the new cloud infrastructure.
- Utilized a strong tech stack including Java, Spring, and Kafka, along with Azure, Azure DevOps, and AWS for cloud services and continuous integration.

Beyondsoft

Guangzhou, China

Backend Engineer (Outsourced to DBS Bank)

Dec. 2021 - Nov. 2022

- Beyondsoft Consulting is a leading business IT and consulting company. DBS Bank is the largest bank in Southeast Asia by assets and among the largest banks in Asia. Contracting with Beyondsoft, assigned to work for DBS Bank and contributed to DBS Client Connect and DBS DigiBank CN projects.
- In the DBS Client Connect project, contributed to the development of the equity trading microservice. Responsibilities included creating functionalities for equities display, customer display, pre-trade checks, and placing actual equity trade orders. Integrated Avaloq APIs to enhance the underlying infrastructure and improved the user experience by implementing an edit distance algorithm to refine stock code searches.
- In the DBS DigiBank CN project, actively contributed to the development of several microservices handling mutual funds management, structured investment products, investment portfolios, and transaction lists. Assisted with performance tests by analyzing logs from Pivotal Cloud Foundry to generate reports on the QPS of the microservices. Developed a tool to automate test generation for the testing tool Karate, streamlining the testing process and improving test coverage.
- Leveraged cloud technologies and modern frameworks, employing automation best practices with tools such as Java, Spring Cloud, Jira, Confluence, Jenkins, Finacle, Pivotal Cloud Foundry, and Kibana, and utilizing BDD and TDD methodologies.

Self-Employed

Guangzhou, China

Full Stack Engineer

Jan. 2020 - Nov. 2021

- Authored and published technical blogs to share knowledge online, improved English proficiency through engaging with Netflix and literature, and enhanced problem-solving skills by tackling around 500 algorithm problems and participating in Codeforces competitions.
- Gained a little hands-on experience with big data and cloud-native technologies by exploring introductory tutorials and running examples for Spark, Hadoop, Kubernetes, and Docker.
- Completed various software projects as a freelancer, including LED Sign Website Development, Enterprise WeChat Integration for ShowMeBug, Web Scraper for Trade Data Collection, and an eBook tool called mathjax2mobi.
- LED Sign Website Development (lvchensign.com): Developed a website for an LED sign manufacturing company using Bootstrap, HTML, and JavaScript. Implemented features for showcasing products.
- Enterprise WeChat Integration for ShowMeBug: Contributed to the integration of ShowMeBug with Enterprise WeChat, enabling seamless access to technical interview tools within the Enterprise WeChat ecosystem. Utilized technologies like Ruby, Ruby On Rails, PostgreSQL, and the WeChat SDK to create a smooth user experience for interviewers and candidates.
- Web Scraper for Trade Data Collection: Developed a web scraper using Python and Selenium to collect trade data for a non-woven fabric company. Automated data extraction and page navigation, processed and stored the data in a SQLite database, and generated reports for business analysis.
- mathjax2mobi: A tool designed to convert HTML content with MathJax equations into eBook-friendly formats. By transforming LaTeX-based MathJax equations into SVG images, it ensures compatibility with eBook formats like MOBI. Technologies used include Python, BeautifulSoup, and Selenium.

Square Root Technology *Beijing, China*

Founder & Full Stack Engineer *Jul. 2016 - Dec. 2019*

- Beijing Square Root Inc. ran two businesses in 3.5 years. From July 2016 to September 2017, it rolled out and operated Fun Live, a knowledge live broadcast platform. From January 2018 to December 2019, it transformed into a software consulting business.
- With Fun Live, users could participate in various knowledge lectures such as programming or design. Users could pay fees to attend live sessions or reward lecturers. The lecturer used the OBS tool to push live streams to the server. Users could participate in the lecture in real-time or see the playback later. The platform integrated seamlessly with WeChat for notifications. Hosted around 80 lectures, gained 30,000 users and millions of page views. From an engineering perspective, performed most of the software development, resulting in around 2,000 code commits across 3 web client apps and 1 server, later open-sourced on GitHub. Leveraged PHP, Vue, HTML, CodeIgniter, MySQL, Redis, LeanCloud, Alibaba Cloud, WeChat SDK, SRS (Simple Realtime Server), WeChat MiniProgram, Fabric, Linux, Nginx, and Sketch.
- During the software consulting business period, completed 50 small software projects for customers, including websites, games, and apps. The revenue was around 3 million RMB and the profit was around 700 thousand RMB. Responsible for project negotiation, team management, and some software development. Notable projects are shown below.
- MianbaoLive: Led a comprehensive backend refactoring effort for MianbaoLive, a one-stop content monetization and social economy platform. Optimized the performance, stability, and user experience of the entire tech stack. Previously used ThinkPHP, Node.js, and Go, and rewrote everything in Laravel for server-side development. The platform included modules for courses, users, content, user attendance, payments, and distribution sales. Partnered with Himalaya, a top audio platform in China, and synchronized content between the platforms.
- Jiangsu TV's Super Brain Program WeChat Mini Program: Responsible for all backend development and half of the frontend development for the Super Brain program's WeChat Mini Program. Engaged viewers with interactive puzzle games, enabling them to compete and rank to become the "Super Brain." Used the WeChat Mini Program framework and Wepy (Vue.js) to create game components and ranking pages. Integrated RESTful APIs to fetch game data and user information. Performed extensive performance tuning to ensure the system could handle high concurrency, leveraging caching technologies such as Redis.
- Chongding Conference: Spearheaded full-stack engineering for Chongding Conference, a mobile trivia app in China resembling HQ Trivia. Designed and implemented services and APIs to handle live trivia events, user management, and real-time question and answer sessions. Utilized Java and Spring for the backend, with Redis and Kafka for caching and message queuing, Zookeeper for service coordination, and Socket.IO for real-time interactions. Developed the admin panel to help operators control the game. The app supported live streaming, real-time interaction, and robust performance under high traffic conditions. Contributed to technical discussions for using SEI (Supplemental Enhancement Information) to synchronize the timestamp of the live streaming with the trivia game interactions.

CodeReview.cn *Beijing, China*

Cofounder & Full Stack Engineer *Nov. 2015 - Jul. 2016*

- Beijing Dami Entertainment Co. was a company founded by 6 Internet enthusiasts, including me. Launched and operated the platform CodeReview, a professional platform for code review, communication, and sharing. Gained around 3,000 users.
- The platform included functionalities such as user management, code submission and review processes, notification systems, payment integration, and event and workshop management. Engineers could submit their code for expert review to improve its quality, and experts received fees for their reviews. The platform also offered workshops and events open to users.
- Responsible for the development of the backend and half of the frontend. Utilized a strong tech stack including PHP, Vue, CodeIgniter, Alibaba Cloud, and Ping++.

LeanCloud

Beijing, China

Software Engineer

Jul. 2014 - Nov. 2015

- Delicious Bookmarks is a leading cloud computing provider in China, known as LeanCloud. It offers a complete set of cloud services including object storage, file storage, web hosting, containers, instant messaging, push notifications, SMS, and game backends. The company serves hundreds of thousands of developer users.
- Participated in the development of LeanCloud Objective-C SDK and Java SDK. Responsible for the development of LeanChat iOS client and Android client, a chat application designed to demonstrate the instant messaging SDK. Additionally, worked on various frontend projects.
- Employed a robust tech stack, leveraging modern tools such as iOS SDK, Android SDK, Cocoapods, Xcode, Android Studio, and the Angular framework.

Education

California Science and Technology University

Milpitas, CA, USA (Online)

Master of Science in Computer Systems and Engineering (MSCSE)

2026.10 - Present

- Pursuing the online Master of Science in Computer Systems and Engineering (MSCSE) at California Science and Technology University (CSTU), Milpitas, CA. The online master program is currently enrolling new students.
- A 12-month online program (30 semester credits: core courses, electives, and capstone) covering full stack development with AI assistance, Python for AI, machine learning, deep learning with PyTorch, cloud computing and security, LLM applications, and generative AI. Graduates may pursue roles such as software engineer, data analyst, machine learning engineer, or AI engineer.

Guangdong University of Foreign Studies

Guangzhou, China

Associate Degree in Computer Application

2022.10 - Present

- This is China's National Higher Education Self-study Examination. The program requires 18 written exams and 5 practical exams, taken 2–3 times per year at organized onsite sessions with proctor teachers.
- Passed 13 written courses: Introduction to Operating Systems, Introduction to Data Structures, Advanced Language Programming (I), English (I), Advanced Mathematics, Database and Applications, Computer Application Technology, Introduction to Law, Computer Organization and Architecture, Introduction to Programming—Java, Mobile Application Development Technology, Web Design and Production, and Ideological and Moral Cultivation and Rule of Law.
- Passed 3 practical exams: Fundamentals of Computer Programming, Computer Fundamentals and Application Technology, and Database and Its Applications.
- Pending 5 written exams: Computer Network Technology, Switching and Routing Technologies, College Chinese, An Overview of Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, and Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics.
- Pending 2 practical exams: Web Design and Production, Mobile Application Development Technology.

Beijing Forestry University

Beijing, China

Bachelor's in Digital Media Arts (Dropout)

2013.09 - 2014.06

- Completed 13 courses, including Computer Introduction, Fundamentals of Programming, Programming Fundamentals Lab, College English, Advanced Mathematics, Principles and Techniques of Digital Imaging, and Two-Dimensional Graphics Technology.
- Built a campus assistant Android application "Bjfu Assistant" that integrates the functions of checking grades, and alumni forums. It has been downloaded and used by thousands of users on campus. And I built an Android App for learning English by watching movie clips, and won the third prize in a hackathon.
- 2014 Blue Bridge Cup Competition First Prize. Ranked first in one ACM/ICPC monthly training competition among approximately 100 top students at the university, including higher-grade students.
- Awarded third prize in the "Youth, Health, New Media" digital works creativity contest organized by UNESCO and the Communication University of China for the digital work "Mom Shy To Say."

Guangzhou Yuyan Middle School

Guangzhou, China

Middle School Diploma, Science Track

July 2007 - June 2013

- Achieved a score of 756 out of 810, ranking in the top 300 out of around 100,000 students in the Guangzhou High School Entrance exam.
- Won first prize among Guangzhou contestants in the first round of the 2011 Guangdong NOIP (National Olympiad in Informatics in Provinces); advanced to the second round without a prize; likely ranked in the top 300.

Corporate Projects

Full Stack Development, Backend Development, Platform Integration, Team Management

Various Corporate Projects *2014 - 2024*

- Note: The major corporate projects are already detailed in the Work Experience section of this resume. The following is a short list of the most significant ones with simple details.
- HSBC Finance Transformation Platform (2025.02 - Present) Participated in the development as a Full Stack Engineer, utilizing tech stacks like Java, Spring, Python, IBM WebSphere, and Copilot.
- AI-powered Story Bot: Full stack development using Claude's API for personalized stories. Technologies: Python, Flask, React, AWS. (May 2024 - Jul. 2024)
- PayMe from HSBC: Backend development for Auto Top Up feature and cloud migration. Technologies: Java, Spring Cloud, Azure, AWS. (Nov. 2022 - Jul. 2023)
- DBS Digibank CN: Backend development for mutual funds microservices. Technologies: Java, Spring Cloud, Pivotal Cloud Foundry. (Jul. 2022 - Nov. 2022)
- DBS Client Connect: Backend development for equity trading microservices. Technologies: Java, Spring Cloud, Pivotal Cloud Foundry. (Dec. 2021 - Jun. 2022)
- ShowMeBug: Platform integration with Enterprise WeChat, including login and interview entrance. Technologies: Ruby on Rails, PostgreSQL, WeChat SDK. (Jul. 2021 - Sep. 2021)
- Square Root Inc Projects: Led 50 projects including websites, games, apps, and WeChat mini programs. Responsibilities included team management, contract negotiation, project management, client communications, and software development. (Jan. 2018 - Dec. 2019)
- Super Brain TV Program WeChat Mini Program: Backend and frontend development for the TV show's WeChat Mini Program. Technologies: Java, Spring, JavaScript, HTML, AliCloud. (Feb. 2018 - Mar. 2018)
- Chongding Conference: Backend and frontend development for live-streaming trivia app. Technologies: Java, Spring, WebSocket, Netty. (Sep. 2017 - Jan. 2018)
- Mianbao Live: Backend refactoring for a content monetization platform. Technologies: PHP, TypeScript, Laravel, Go. (Sep. 2017 - Jan. 2018)
- Fun Live: Full stack development for a knowledge live broadcast platform. Technologies: PHP, JavaScript, HTML, AliCloud, RTMP Streaming. (Jun. 2016 - Sep. 2017)
- LeanCloud SDK and LeanChat: Development of Objective-C and Java SDKs and their demo app LeanChat. (Jul. 2014 - Oct. 2015)

Personal Projects

Model Training, Personal AI Tools, Open Source Contributions

AI Projects *2023 - Present*

- Trained GPT-2 124M from scratch twice using nanoGPT on the FineWeb dataset —once on a RunPod H200 (141 GB HBM3e), once on a DigitalOcean H100 (80 GB HBM3); also ran experiments on a home RTX 4070 (12 GB). Forked Karpathy's nanoGPT with 45 commits: FineWeb shard-based pipelines, inline tensor shape annotations, and training configs scaled for different GPU budgets. Tech: Python, PyTorch, CUDA, HuggingFace Datasets, tiktoken.
- Built nanoGPT fork with 6 dataset pipelines (FineWeb-Edu 10B+ tokens, OpenWebText, Wikipedia) and 6 training configs (124M–1.5B params). Also trained GPT-2 760M on AMD MI300X (192 GB HBM3) via AMD Dev Cloud. Tech: Python, PyTorch, ROCm, FlashAttention, HuggingFace Hub.
- MoE & LLM Inference: Expand from training into MoE & serving systems by reading code, running it, and compiling from source. Studied MoE end-to-end (DeepSeek-MoE, MegaBlocks, Tutel, Mixtral) —routing, dispatch, expert compute, combine, load balancing. Ran nano-vLLM on an RTX 4070 serving Qwen3-0.6B with flash-attention (506 tok/s prefill, ~1434 tok/s benchmark) and learned the serving stack: paged KV cache, chunked prefill/decode, continuous batching, CUDA graphs, tensor parallelism. Compiled SGLang from source (built its 3 PyO3 Rust extensions) serving Qwen2.5-0.5B over an OpenAI-compatible API. Tech: Python, PyTorch, CUDA, Triton, FlashAttention, nano-vLLM, SGLang, vLLM, ROCm, NCCL.
- SEC-EDGAR-GPT: Trained GPT-2 124M from scratch on 1.55B tokens of SEC EDGAR financial filings (10-K, 10-Q, 8-K) using nanoGPT on a single RTX 4070 (12 GB). Converged to validation loss 2.28 in ~8 hours. Built a paper, project website (Cloudflare Workers), chat UI (nanoGPT server), and HuggingFace model repo. Whole project done in 3 days via Hermes Agent. Tech: Python, PyTorch, nanoGPT, HuggingFace Hub, Cloudflare Workers, tiktoken.
- zz: Dataset processing toolkit —resumable shard-based download with token budget planning, memory-safe parquet extraction via pyarrow, tokenization, training log analysis. Hosted on GitHub and Hugging Face. Tech: Python, pyarrow, tiktoken, pandas, matplotlib.
- Consumed ~5B tokens across platforms: OpenRouter 2.28B tokens ($239, 155K requests), DeepSeek 629M tokens, Claude API 115M+ tokens, Xiaomi MIMO 1.25B tokens. Tech: OpenRouter API, Claude API, DeepSeek, Copilot API, REST APIs.
- jekyll-ai-blog: AI-powered blog with LLM-based multi-language translation (9+ languages), Google Cloud TTS, XeLaTeX PDF/EPUB, and GitHub Actions CI/CD. 10K+ English posts, 10K+ Chinese posts, 9.7K AI notes. Tech: Jekyll, Python, Google Cloud TTS, XeLaTeX, DeepSeek, Mistral.
- lzwjava.com: Personal blog with ~400 posts and ~8,000 AI answer notes; ~70K page views/month (Cloudflare). Tech: Jekyll, GitHub Pages, Cloudflare Analytics, MathJax.
- ww: Cross-platform CLI toolkit (255+ commits, 10+ command groups) —git workflows with AI commit messages, note management, image/PDF processing, Copilot chat, AMD Dev Cloud management. Tech: Python, uv, Gemini Flash, OpenRouter, Pillow.
- FluxReel: AMD AI DevMaster Hackathon 2026-07 (Track 1) —turns a one-line topic into a 15-second vertical short video (1080×1920, 30 fps) with image generation on an AMD Radeon GPU (ROCm). Pipeline: LLM script + 5 bilingual scenes, parallel image gen (diffusers FLUX.1-schnell / stable-diffusion.cpp GGUF / OpenRouter fallback), PIL slide composition with CJK captions, ffmpeg assembly. Includes FastAPI web UI + REST API, optional YouTube upload, remote AMD GPU management. Earned the AMD AI DevMaster Hackathon Completion Award. Tech: Python, diffusers, FLUX.1-schnell, stable-diffusion.cpp, ffmpeg, PIL, FastAPI, ROCm, OpenRouter.
- iclaw: Terminal AI agent REPL with native tool calling —codes, searches, runs commands autonomously. Supports Copilot OAuth and OpenRouter; works in locked-down enterprises. Tech: Python, Copilot OAuth, OpenRouter API, DuckDuckGo, Tavily.
- live-server modernization: Used AI agents (Claude Code, Codex) to Dockerize, upgrade CodeIgniter and Vue, modernize full stack of a startup that served 30K users. Tech: PHP, Docker, CodeIgniter, Vue.js, MySQL, Redis.

Open Source Contributions, Blog Development, Backend and Frontend Development, Mobile App Development, Tool Development

Various Personal Projects *2013 - Present*

- Algorithm Solutions: Solutions to algorithm problems, hosted on GitHub with 2466 commits, implemented primarily in Java. (2013 - Present)
- Personal Blog: Maintained a bilingual blog (English and Chinese) with 500 commits, sharing insights on programming, technology, and personal development. (2013 - Present)
- Live Server: Backend development of a knowledge live broadcast platform using PHP with 660 commits. (2016 - 2017)
- Live Mobile Web: Developed the mobile frontend of a knowledge live broadcast platform using Vue and JavaScript with 528 commits. (2016 - 2017)
- Live Web: Developed the desktop frontend of a knowledge live broadcast platform using Vue with 140 commits. (2016 - 2017)
- Live WeChat Mini Program: Developed the WeChat Mini Program for a knowledge live broadcast platform using JavaScript with 63 commits. (2016 - 2017)
- Code Review Server: Developed the backend for a professional platform for code review using PHP with 275 commits. (2015 - 2016)
- Code Review Web: Developed the frontend for a professional platform for code review using Vue and JavaScript with 302 commits. (2015 - 2016)
- LeanChat iOS: Developed an iOS chat application demonstrating LeanCloud SDK, implemented in Objective-C with 556 commits. (2014 - 2015)
- LeanChat Android: Developed an Android chat application demonstrating LeanCloud SDK, implemented in Java with 412 commits. (2014 - 2015)
- Feynman Lectures Mobi: Developed a tool to convert LaTeX to SVG for creating mobi ebooks, implemented in Python with 47 commits. (2020 - 2021)

Blog Posts

Below is a selection of my blog posts, covering personal growth stories, startup journeys, English learning, programming, geek projects, and more.

Selected Blog Posts

- LLM Costs, Agents, and Coding Tools
- Optimizing Software Engineering for AI
- Building a Scalable Application on Azure
- Comprehensive Guide to Spring Framework
- Build an AI-Powered Story Bot with Flask, React, and ELK
- Deploying with Pivotal Cloud Foundry
- The Wisdom of Learning
- Zen and the Art of Machine Learning
- How to Build a Computer
- Recommended Blog Posts for Engineers
- Meta-Thinking Framework
- Complete Life Philosophy
- Comfortable Daily Eyeglass Use at 1.50D Below Full Prescription
- Complete Guide to Learning English
- Setting Up Live Streaming with OBS, SRS, and FFmpeg
- Lieyunwang: Post-95 Entrepreneur Founded Knowledge Live Streaming Platform QuZhibo
- Enhancing iOS Development with Automated Testing and Tools
- My Trip to the United States: First Impressions of Silicon Valley
- Porting FFmpeg to Android
- A Deep Dive into Custom Drawing in Android

Papers

I'm a self-taught researcher with two academic-style papers —one on reversing myopia and natural vision restoration, inspired by the works of Yin Wang and Todd Becker, supported by three years of experimentation, and one on training GPT-2 from scratch on SEC filings as an AI research contribution. In computer science, I am still striving to make similar breakthroughs.

Natural Vision Restoration: Experimental Verification, Astigmatism Reduction, and the Just Barely Clear Principle — *Zhiwei's Blog*

Li, Zhiwei — *Aug. 2026*

- Combined three papers into one unified study verifying a natural vision restoration method, analyzing astigmatism reduction during myopia reversal, and introducing the just barely clear principle.
- Published online: https://lzwjava.com/natural-vision-restoration-en

SEC-EDGAR-GPT: A GPT-2 (124M) Language Model Trained from Scratch on SEC EDGAR Filings — *GitHub*

Li, Zhiwei — *Jun. 2026*

- Trained a 124M-parameter GPT-2 from scratch on 1.55B tokens of SEC EDGAR financial filings (10-K, 10-Q, 8-K) using nanoGPT on a single RTX 4070. Converged to validation loss 2.28 in extasciitilde8 hours.
- Published online: sec-edgar-gpt.pdf
- Code: github.com/lzwjava/sec-edgar-gpt | Model: huggingface.co/lzwjava/sec-edgar-gpt-124m-hf

Skills

- Started programming in 2008. The following shows relative familiarity by cumulative usage time, distinct from work experience.
- 5+ Years of Experience: Chinese, English, Writing, Java, Algorithm, Test-driven Development, Cloud Computing, DevOps, Backend Development, AI-assisted Development
- 3-4 Years of Experience: Cantonese, iOS Development, Android Development, Machine Learning, LLM APIs, AI Agent Development
- 1-2 Years of Experience: Product Management, Frontend Development, GPU Training and Inference (PyTorch, CUDA, ROCm), Prompt Engineering, Model Fine-tuning
- Less than 1 Year of Experience: Japanese

Tools

Below are detailed tools that help me to accumulate the above skills, plus other interesting tools. I am still using some of them and can learn for a few days to use the rest of them again.

- Natural Language: English, Chinese, Cantonese, Hakka, Japanese
- Operating System: Linux, Windows, MacOS, Android, iOS, OpenWRT
- Programming Language: PHP, Java, Objective-C, Swift, Ruby, HTML, C, C++, JavaScript, Lisp, Clojure, TypeScript, Node.js, Go, Python, Pascal
- Java: JDK 8/17, Spring MVC/Core, SLF4J, Jackson, Hibernate, JaCoCo, FeignClient, XML, Reflections, Servlet, Executor
- Python: 3.8/3.11, Flask, logging, json, SQLAlchemy, unittest, requests, xml.etree, inspect, http.server, concurrent.futures
- Domain-specific language: JSON, CSS, SQL, NoSQL, Kusto Query Language, Kibana Query Language
- Database: MySQL, SQLite, PostgreSQL, Microsoft SQL Server, Amazon RDS, MongoDB, IBM Db2, Navicat
- Database Client: SQuirreL SQL Client, DBeaver, MySQL Workbench, Sequel Pro, Navicat, Eclipse Data Source Explorer
- Frontend: Vue, React, Angular, jQuery, Hexo, Jekyll, Stylus, Sass, Chrome, Safari
- Framework: RTMP streaming, HTTP/IP, Spring, Spring Boot, Hibernate, Netty, CodeIgniter, Laravel, Flask, Express, Socket.IO, iOS SDK, Android SDK, Flutter
- Big Data: Redis, Kafka, Spark, Flink, ZooKeeper
- Math: Calculus, Linear Algebra, Discrete mathematics, Latex
- IDE: Emacs, IntelliJ IDEA, Visual Studio Code, Xcode, Android Studio, Eclipse, PyCharm, PhpStorm, Sublime Text, Atom, WebStorm, Delphi, Turbo Pascal
- Terminal: Powershell, Command Prompt, iTerm2, GNOME Terminal, Terminal (macOS)
- Development: NPM, Webpack, Git, HomeBrew, Gradle, Maven, Sketch, Cocopods, WordPress, YourKit Java Profiler
- CI/CD: Travis CI, GitHub Actions, Jenkins, Azure DevOps, Control-M
- Monitoring: SonarQube, Prometheus, Elasticsearch, Logstash, Kibana, Nginx, Docker, Kubernetes
- Machine Learning: TensorFlow, PyTorch, Theano, Anaconda, Jupyter Notebook, CUDA, Numpy, Pandas
- Cloud: GitHub, GitLab, Digital Ocean, Vultr, AWS, Azure, Google Cloud Platform, Linode, Alibaba Cloud, Qiniu Cloud, Tencent Cloud, LeanCloud, Firebase, Cloudflare, GoDaddy, Agora, Vercel
- Application Runtime: IBM WebSphere Liberty Server, IBM WebSphere Application Server, Pivotal Cloud Foundry, PM2
- Project: Jira, Confluence, ServiceNow, Trello, Microsoft To Do, Tower.im
- Large Language Model: OpenAI, Claude, MiniMax, Amazon Bedrock, Copilot, LangChain, vLLM, llama.cpp, OpenRouter, Claude Code, Codex, Copilot API, Claude Code Router
- Linux: nmap, grep, awk, ffmpeg, htop, df, lsof, strace, tcpdump, iostat, iptables, rsync, journalctl, systemctl, curl, tmux, traceroute, emacs, diff
- Testing: Postman, Selenium, Karate, JUnit, Spring Boot Starter Test, Mockito, Apache JMeter, Apache ab, Playwright, Python unittest
- Hardware: Arduino UNO, Raspberry Pi Pico

Knowledge

Below is a summary of my expertise and knowledge domains. While I have a solid understanding of some topics and a more basic familiarity with others, I would be happy to share more in an interview if you are interested in any specific areas.

Industry Knowledge

- Banking and Payments, Education and E-learning Platforms, Social Media and Content Platforms, User and Account Management, Notification Systems, Data and Analytics, Mobile Applications, Payment Systems, Transaction Management, Social Media Integration, Video Management, Reward Systems, Live View Tracking, Application Management, Mutual Funds and Equity Trading, Browser-Based Programming Environments.

Computer Science Fundamentals

- Advanced Mathematics, Computer Organization, Operating Systems, Computer Network Technology, Database and Applications, Computer Application Technology, Data Structures and Algorithms, Microcomputer and Interface Technology.

Technology and Development

- Multilingual Communication, Cross-Platform Development, Full-Stack Programming, Database Management, Machine Learning and Big Data, Mathematical Proficiency, Development Tools, Machine Learning Implementation, Advanced Linux Usage, Testing and Quality Assurance, API Integration, Open Source Contributions, Technical Writing and Blogging, Cloud Computing Services, Distributed Systems, High Performance Optimization, Realtime Application, RTMP Streaming.
- Continuous Integration/Continuous Deployment (CI/CD), Containerization and Orchestration, Network Security, Agile Methodologies, Software Architecture, DevOps Practices, Cloud-Native Applications, API Development, Version Control Systems, Serverless Computing, Performance Monitoring, Data Engineering, Security Best Practices, Software Development Lifecycle (SDLC), Technical Mentoring, Project Management.

Entrepreneurial Spirit and Product Mindset

- Founded two startups: one with co-founders, and one created by myself. The latter gained 30k users and once achieved revenue of 3 million CNY a year.
- Secured an investment of around half a million CNY in my startup at 22 years old, and returned it to the investor when shutting down.
- Applied to Y Combinator twice. One time, entered the second round with around a 3% acceptance rate, though didn't make it into the batch.
- Like to initiate new ideas in big companies to actually help the project; tried many ways and created proofs-of-concept in local development in some code branches. Presented them to tech leads to move things forward. Some received good feedback.
- Gained a lot of social media influencing practice, including blogging, WeChat Subscription channel, TikTok, WeChat Videos, Weibo and LinkedIn. Acquired thousands of followers or 60k page views a year.
- Used 3 mobile phones and thousands of apps and games to experiment with, including different kinds of apps from tens of countries.
- As an early user of Blue Sky, I commented on Jeff Dean's first post here and got a like from him; he is Google's Chief Scientist.
- Own hundreds of small tools and gadgets, such as an Electric/Magnetic Fields meter, etc.
- If things or products are cheap, very willing to buy different kinds to compare and understand their nuances.
- Read around 100 books about business, management, and product. Have some connections with people who are experts in these areas.
- As a crypto and US stock investor since 2018, I took a break due to some losses but resumed investing in 2024, using long-term and regular investing practices. Despite the modest amount, I am pleased to have gained a 25% profit from NVIDIA after one year of regular investing as of July 2025, as well as a 30% profit from some of my Bitcoin investments.

Certificates

2014 **Beijing Forestry University Dropout Certificate**, Beijing Forestry University

2014 **China Higher Education Official Report (Bachelor Education, Dropout Status)**, China Higher Education Bureau

2025 **Guangdong Province Higher Education Self-Study Examination Transcript (13 Subjects Passed)**, Guangdong Province Higher Education Bureau

2026 **IELTS Academics, Score 6.5**, British Council

2023 **Machine Learning Specialization Course Completion Certificate**, DeepLearning.AI and Stanford University

2023 **Deep Learning Specialization Course Completion Certificate**, DeepLearning.AI

2023 **Developing on AWS**, AWS Training

2026 **AMD AI DevMaster Hackathon Completion Award (FluxReel)**, AMD

Links

- **GitHub:** https://github.com/lzwjava
- **Blog:** https://lzwjava.com
- **AI Portfolio:** https://lzwjava.com/ai-portfolio-en
- **Software Portfolio:** https://lzwjava.com/portfolio-en
- **Codeforces:** https://codeforces.com/profile/lzwjava
- **UVa Online Judge:** https://uhunt.onlinejudge.org/id/113519
- **Kaggle:** https://www.kaggle.com/lzwjava
- **LinkedIn:** https://www.linkedin.com/in/lzwjava